

Alive Yi: Interactive Preservation of Yi Minority Embroidery Patterns through Digital Innovation

Zhiwei Wang
Southwest Minzu University
China
21300185@swun.edu.cn

Kexin Nie
The University of Sydney
Australia
niekexinbella@gmail.com

Yuzhe Xia
Southwest Minzu University
China
2857942619@qq.com

Mengyao Guo*
Shenzhen International School of Design, Harbin Institute
of Technology
China
University of Macau
China
coco.mengyao@gmail.com



Figure 1: The display of generated Yi Minority embroidery patterns in “Alive Yi”.

Abstract

By integrating heritage, design, and technology, “Alive Yi” innovatively empowers intangible cultural heritage through interactive generative design. Using our extensive pattern database, we visualize traditional Yi minority embroidery by applying particle and 3D effects in TouchDesigner with Leap Motion, emphasizing their symbolic meanings. This interactivity allows users to transform heritage visuals in real-time, fostering deeper engagement and appreciation. Future plans include using AI tools to create designs appealing to youth. Despite technical challenges, this approach promises cultural resilience in the digital age, offering a blueprint for leveraging computational creativity to preserve and sustain intangible culture.

CCS Concepts

• Human-centered computing → Interaction design; Interactive systems and tools; • Applied computing → Media arts.

*Mengyao Guo is the Corresponding Author.

Keywords

Interactive Visualization, Cultural Preservation, Computational Heritage, Generative Patterns Design

ACM Reference Format:

Zhiwei Wang, Yuzhe Xia, Kexin Nie, and Mengyao Guo. 2024. Alive Yi: Interactive Preservation of Yi Minority Embroidery Patterns through Digital Innovation. In *SIGGRAPH Asia 2024 Posters (SA Posters '24)*, December 03–06, 2024, Tokyo, Japan. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3681756.3697919>

1 Introduction

Chinese traditional patterns are integral to the country’s intangible cultural heritage [Dang et al. 2021]. The embroidery of the Yi minority, as a unique cultural expression, carries the legacy of recording history, passing down myths, conveying aesthetic values, and reinforcing traditional beliefs [Xu and Bae 2016]. Despite the Chinese government’s efforts to encourage the inheritance of this cultural heritage, current preservation methods often fail to capture the attention and engagement of the youth. Traditional approaches, such as physical archives or databases, cater to professionals overlooking the general public who are eager to connect with these cultural legacies.

Digital heritage preservation emerges as a promising solution, offering innovative ways to satisfy the curiosity of the youth and facilitate the dissemination of cultural knowledge [Gao et al. 2024]. However, existing efforts often focus on isolated data collection and cataloging without fully harnessing the immersiveness and interactivity of digital tools to foster meaningful interactions and a sense of collective ownership among diverse stakeholders. To



Figure 2: The digital gene library of patterns.

bridge this gap, our Digital Gene Library of Patterns provides a wealth of resources on Yi embroidery patterns. We created an interactive project, “Alive Yi,” by utilizing TouchDesigner and Leap Motion, which empowers wider audiences to interact, preserve and revitalize these precious cultural assets actively.

2 Technical Support

Our *Digital Gene Library of Patterns* is a comprehensive platform that digitizes and preserves the intricate embroidery patterns of the Yi minority. Built on a robust backend infrastructure using FastAPI and PyTorch, this library aims to enhance the digitalization and management of these traditional patterns.

In this library, users can explore and interact with the digital repository in various ways: visualizing pattern data through management pages showing totals, regional distribution, and operational details (see Figure 2a); uploading patterns with metadata, which the library organizes automatically (see Figure 2b); editing uploaded patterns and collection details (see Figure 2c); and retrieving patterns via keyword search or visual similarity (see Figure 2d). The library also maintains scalability for future enhancements like AI-based pattern generation.

3 Design Approach

Since July 2020, twelve researchers and designers have collected around 50,000 embroidery patterns of the Yi minority from Ganluo County in the Liangshan Yi Autonomous Prefecture of Sichuan Province to document this cultural heritage. In 2023, we interviewed Axiwuzhimo and Abumuji, the intangible cultural heritage inheritors of Yi embroidery. They shared the deep cultural and symbolic meanings behind specific embroidery patterns, such as the sheep horn pattern, the rooster comb pattern, the hibiscus pattern, and the camellia pattern, which inspired us to design corresponding interactive experiences for their digital preservation. To prepare the patterns for digitization, we used Adobe Photoshop to improve the original embroidered samples to acquire high-quality digital versions. We then vectorized these patterns using CorelDRAW, allowing us to scale them up or down without losing any details (see Figure 4). Finally, by integrating TouchDesigner with Leap Motion, we transformed the vector patterns into dynamic, interactive digital representations of Yi minority embroidery (see Figure 3).

The generated effects have both particle and 3D visualizations. The particle effects draw insights from interviews with Yi minority embroidery inheritors, and we developed a multi-layered approach that captures and divides the color of the patterns (see Figure 4). By separating the colors within the embroidered patterns and translating them into dynamic particle effects, “Alive Yi” showcases the depth and richness of the cultural heritage embedded in the Yi minority’s textiles, hats, waist belts, and cultural artifacts (see

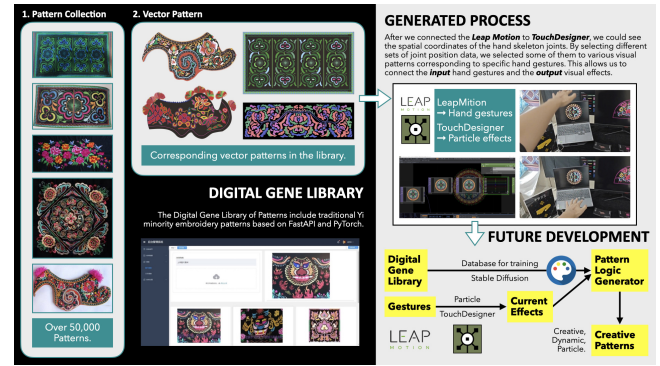


Figure 3: The workflow of “Alive Yi”.

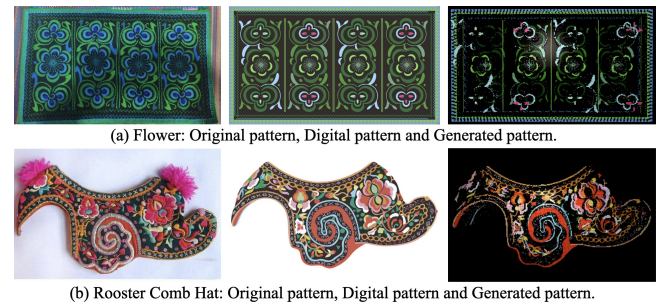


Figure 4: The comparison between the original embroidery patterns, vector patterns, and generated patterns.

Figure 1). Integrating 3-D visualizations further enhances the interactive experience, allowing users to engage with the patterns in real-time, responding to hand gestures and movements.

4 Conclusion

“Alive Yi” presents an interactive digital method for generating and appreciating traditional Chinese patterns using TouchDesigner with Leap Motion. It highlights the symbolic cultural meanings through dynamic 3D transformations of Yi minority motifs. In the next phase, we plan to integrate Stable Diffusion’s API with the current project to create new composite patterns, enhancing youth engagement and encouraging the appreciation of intangible cultural heritage (see Figure 3). The Digital Gene Library facilitates this process, but we face technical complexities in integrating AI and achieving real-time interactivity. We plan to develop it as a digital heritage resource for remixing creative assets from diverse ethnic groups, incorporating contemporary design elements to maintain relevance for youth.

References

- Qiong Dang, Zhongming Luo, Chuhao Ouyang, Lin Wang, and Mei Xie. 2021. Intangible cultural heritage in China: a visual analysis of research hotspots, frontiers, and trends using citeSpace. *Sustainability* 13, 17 (2021), 9865.
- Ze Gao, Mengyao Guo, Yutong Chen, Dongliang Xu, and Zhiwei Wang. 2024. The Digital Blossoming: Generative Flowers of Ethnic Wisdom. In *Proceedings of the 16th Conference on Creativity & Cognition*. 532–536.
- Rui Xu and Soojeong Bae. 2016. An analysis on the form and decoration of Chinese minority women’s waistbands. *Journal of Fashion Business* 20, 3 (2016), 68–88.